```
"""
STEP 0: Install required libraries for RAG pipeline

This step ensures all required packages are available.
Even if already installed in Colab, we include this for completeness.
"""

# Install LangChain (for building RAG pipeline)
!pip install langchain

# Install ChromaDB (vector database)
!pip install chromadb

# Install sentence-transformers (for embeddings)
!pip install sentence-transformers

# Install OpenAI (for LLM)
!pip install openai
```

```
Successfully installed bcrypt-5.0.0 build-1.4.2 chromadb-1.5.7 durationpy-0.10 kubernetes-35.0.0 onnxruntime-1.24.4 opentelemet
Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
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Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20.0
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.20
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Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch>=1.11.0
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Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from openai) (1.9.0)
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Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->opena
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3,>=1.9.0->op
```

```
"""
```

```
Install updated LangChain ecosystem packages
"""

!pip install langchain langchain-community langchain-openai chromadb sentence-transformers langchain-huggingface -q
```

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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 88.5/88.5 kB 4.8 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.0/1.0 MB 23.0 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 64.9/64.9 kB 4.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 51.0/51.0 kB 1.7 MB/s eta 0:00:00
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.33.1 which is incompatible.
google-adk 1.28.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.0 which is incompatible.
google-adk 1.28.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.0 which is incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have openteleme
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto 1.40.0 whic
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.40.0 which is
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.40.0 wh
```

```
from langchain_huggingface.embeddings import HuggingFaceEmbeddings

"""
STEP 2: Load embedding model

Embeddings convert text into numerical vectors.
We use a pre-trained model: all-MiniLM-L6-v2
"""

# Initialize embedding model
embedding_model = HuggingFaceEmbeddings(
    model_name="all-MiniLM-L6-v2"
)

print("Embedding model loaded successfully")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 35.5kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 10.3kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
README.md: 10.5k/? [00:00<00:00, 674kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 4.29kB/s]
config.json: 100%  612/612 [00:00<00:00, 48.1kB/s]
model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 454MB/s]
Loading weights: 100%  103/103 [00:00<00:00, 438.68it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%  350/350 [00:00<00:00, 27.4kB/s]
vocab.txt: 232k/? [00:00<00:00, 10.3MB/s]
tokenizer.json: 466k/? [00:00<00:00, 19.4MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 5.25kB/s]
config.json: 100%  190/190 [00:00<00:00, 18.2kB/s]
Embedding model loaded successfully
```

```
"""
Remove old/broken LangChain installations
"""
!pip uninstall -y langchain langchain-community langchain-openai
```

```
Found existing installation: langchain 1.2.14
Uninstalling langchain-1.2.14:
  Successfully uninstalled langchain-1.2.14
Found existing installation: langchain-community 0.4.1
Uninstalling langchain-community-0.4.1:
  Successfully uninstalled langchain-community-0.4.1
Found existing installation: langchain-openai 1.1.12
Uninstalling langchain-openai-1.1.12:
  Successfully uninstalled langchain-openai-1.1.12
```

```
"""
STEP 1: Completely remove conflicting packages
"""

!pip uninstall -y langchain langchain-core langchain-community langchain-openai langchain-text-splitters
```

```
WARNING: Skipping langchain as it is not installed.
Found existing installation: langchain-core 1.2.23
Uninstalling langchain-core-1.2.23:
  Successfully uninstalled langchain-core-1.2.23
WARNING: Skipping langchain-community as it is not installed.
WARNING: Skipping langchain-openai as it is not installed.
Found existing installation: langchain-text-splitters 1.1.1
Uninstalling langchain-text-splitters-1.1.1:
  Successfully uninstalled langchain-text-splitters-1.1.1
```

```
"""
Install required packages (compatible with current environment)
"""

!pip install langchain chromadb sentence-transformers -q
```

```
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   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 508.7/508.7 kB 12.1 MB/s eta 0:00:00
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 169.8/169.8 kB 5.8 MB/s eta 0:00:00
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the
langchain-classic 1.0.3 requires langchain-text-splitters<2.0.0,>=1.1.1, which is not installed.
```

```
"""
Install missing dependency
"""
!pip install langchain-text-splitters -q
```

```
!pip install chromadb sentence-transformers -q
```

```
"""
Import required libraries
"""

from sentence_transformers import SentenceTransformer
import chromadb
```

```
"""
Load embedding model
"""

model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
Loading weights: 100%|| 103/103 [00:00<00:00, 381.29it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

```
"""
Sample dataset
"""

documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
```

```
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]
```

```
"""
Create ChromaDB client and collection
"""

client = chromadb.Client()

collection = client.create_collection(name="rag_collection")
```

```
"""
Convert documents into embeddings and store
"""

for i, doc in enumerate(documents):
    embedding = model.encode(doc).tolist()

    collection.add(
        ids=[str(i)],
        documents=[doc],
        embeddings=[embedding]
    )
```

```
"""
Ask a query
"""

query = "What is Artificial Intelligence?"

query_embedding = model.encode(query).tolist()
```

```
"""
Retrieve most relevant documents
"""

results = collection.query(
    query_embeddings=[query_embedding],
    n_results=2
)

print("Retrieved Documents:")
for doc in results['documents'][0]:
    print("-", doc)
```

```
Retrieved Documents:
- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
```

```
"""
Generate answer (basic RAG output)
"""

context = " ".join(results['documents'][0])

answer = f"Based on retrieved data: {context}"

print("\nFinal Answer:")
print(answer)
```

```

Final Answer:
Based on retrieved data: Artificial Intelligence is the simulation of human intelligence. Machine Learning is a subset of AI.
```